

CIRCUIT TO PROTECT APPLIANCES From High/Low Voltages

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Voltage fluctuations and power cuts can damage electrical equipment and goods like submersible pumps, air-

conditioners, television sets, and refrigerators. The circuit presented here (see Fig. 1) will protect the costly electrical items from high as

well as low voltages.

The circuit comprises step-down transformer X1, 2N2222 transistors T1 and T2, 1N4007 rectifier diodes D1-D3, a 5mm LED (LED1), 5.6V Zener diodes ZD1 and ZD2, 4.7k potmeters VR1 and VR2, 12V single-changeover relay RL1, and a few other components.

The circuit works on 12V power supply derived from 230V AC primary to 12V-0-12V, 500mA secondary step-down transformer X1. Diodes D1 and D2 rectify the AC into pulsating DC, which is smoothened by the 220 μ F, 35V filter capacitor C1 to produce a regulated 12V DC voltage to operate the circuit.

This 12V voltage increases or decreases slightly based on higher or lower than the normal 230V supply of AC mains input received through transformer X1.

When the AC mains voltage is between the normally-acceptable range of 190V and 230V, the DC voltage at cathode of Zener diode ZD1 is less than 5.6V. As a result, transistor T1 is non-conductive. At the same time, DC voltage at the cathode of Zener

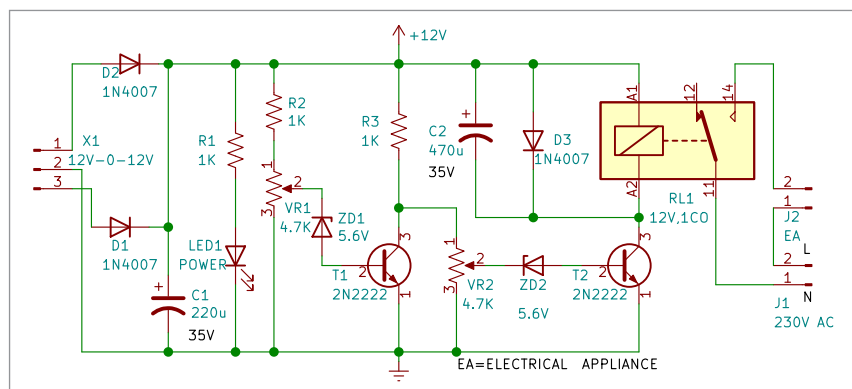


Fig. 1: Circuit diagram

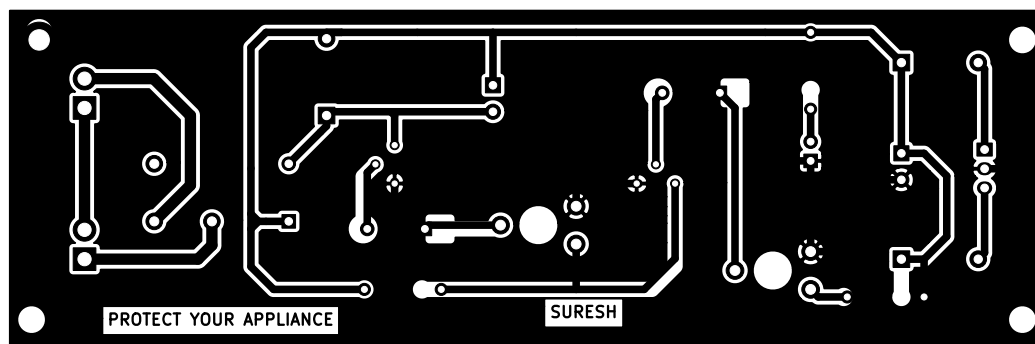


Fig. 2: Actual-size PCB layout

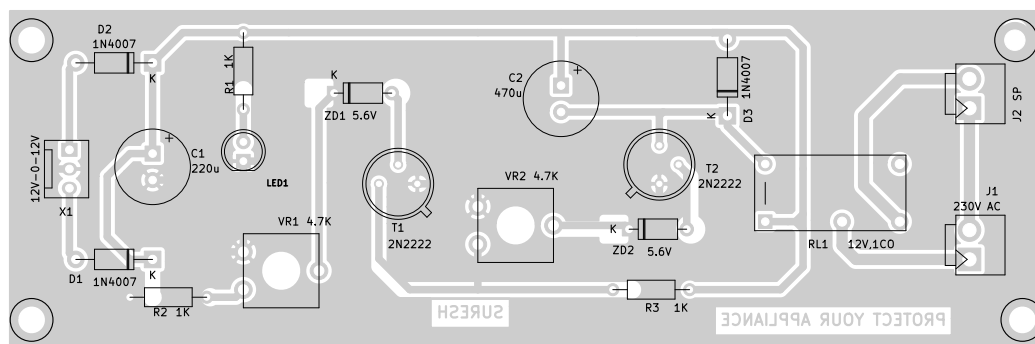


Fig. 3: Component layout of the PCB